

Student Name:

Daytime: ☐Evening: ☐

Student No :

(25)1-Let $x=2^5+2^{-10}+2^{-19}$ Find the machine numbers on the Marc-32 that are just to the right and just to the left of x .
 (Marc-32 number system: $x=(1.a_1a_2a_3 \dots a_{22}a_{23})_2 * 2^m$ where $-126 \leq m \leq 127$)

(25)3-Let $f(x)=x^3+2x^2-5x-6$. Perform one iteration of Secant method on $f(x)$ with starting points $x_0=1$ and $x_1=-1$ (Compute x_2)

(30)2- Let $x=0.643676895$, and $y=0.643671734$.

- Consider computing $x+y$ using a decimal computer having six-digit mantissa. What will be the relative error in percent?
- Consider computing $x-y$ using a decimal computer having six-digit mantissa. What will be the relative error in percent?

(25)4-Let $f(x)=x^3+2x^2-5x-6$. Perform two iterations of Newton's method on $f(x)$ with starting point $x_0=1$. (compute x_1 , and x_2)